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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

CV2336-1- HYDROLOGY AND IRRIGATION ENGINEERING
(For CV)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Hydrology is important in civil engineering for:
A) Flood control
B) Irrigation planning
C) Hydropower design
D) All of the above
2. The majority of Earth's water is stored in:
A) Oceans
B) Rivers
C) Lakes
D) Glaciers
3. Thiessen polygon method is used for:
A) Runoff estimation
B) Mean rainfall computation
C) Groundwater modeling
D) Evapotranspiration estimation
4. Infiltration index represents:
A) Effective rainfall
B) Soil moisture
C) Storage capacity
D) Evaporation loss
5. Factor that increases evaporation is:
A) High wind speed
B) High humidity
C) Dense vegetation
D) Low temperature
6. Runoff mainly depends on:
A) Rainfall intensity
B) Land use
C) Soil type
D) All of the above
7. Rational formula for peak runoff includes coefficient (C), which represents:
A) Rainfall intensity
B) Runoff coefficient
C) Catchment area
D) Soil infiltration
8. Rainfall mass curve is a plot of:
A) Cumulative rainfall vs. time
B) Rainfall intensity vs. time
C) Runoff vs. time
D) Effective rainfall vs. area
9. The major ill-effect of water logging is:
A) Increase in soil aeration
B) Reduction in crop yield
C) Improvement in drainage
D) Decrease in soil salinity
10. Sprinkler irrigation is best suited for:
A) Sandy soils
B) Heavy clay soils
C) Paddy fields
D) Orchards only
11. The irrigation method most efficient in saving water is:
A) Flood irrigation
B) Sprinkler irrigation
C) Drip irrigation
D) Furrow irrigation
12. Irrigation frequency is influenced by:
A) Soil water holding capacity
B) Canal alignment
C) Seepage losses
D) Type of regulator

13. Irrigation efficiency generally ranges between:
 A) 20–30% B) 30–40%
 C) 40–50% D) 50–70%
14. A canal drop is a structure constructed to:
 A) Raise the water level B) Control waterlogging
 C) Dissipate surplus energy D) Improve irrigation efficiency
15. Cross drainage works are required when:
 A) Canal crosses a road B) Canal crosses a river or drain
 C) Canal crosses a field boundary D) Canal crosses a contour line
16. According to Lacey's theory, silt factor is a function of:
 A) Velocity B) Size of sediment particles
 C) Discharge D) Canal depth
17. The storage zone between dead storage and full reservoir level is called:
 A) Surge storage B) Live storage
 C) Dead storage D) Reservoir bed
18. A gravity dam fails by overturning when:
 A) Resultant force passes outside the middle third B) Water level is low
 C) Uplift pressure is zero D) Dam is too short
19. In stability analysis of gravity dams, the main resisting force is:
 A) Water pressure B) Uplift pressure
 C) Self-weight of dam D) Seismic force
20. The most common cause of hydraulic failure in earthen dams is:
 A) Piping through the body or foundation B) Overheating of soil
 C) Excessive vegetation D) Poor compaction of crest only

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List any four forms of precipitation and Explain orographic precipitation with a neat sketch.	8	L*2	1	1
	b) Define evapotranspiration. Differentiate between PET and AET.	8	L2	2	2
2.	a) Explain various methods for estimating evapotranspiration.	8	L2	1	1
	b) Write notes on components of runoff and the factors influencing runoff.	8	L2	2	2
3.	a) The catchment is in the form of circle of radius 10 km. There are five rain gauge stations. One station (E) is located at the center of circular area with coordinates(0,0) and the other four stations A,B,C,D are located at (5,5),(-5,5),(-5,-5) and (5,-5) respectively. The rainfall recorded at these stations are A(8cm),B(9cm),C(7cm), D(10 cm) and E(12) cm respectively. Determine the average depth of rainfall over the catchment using Thiessen polygon method. Compare this with arithmetic mean method.	8	L3	2	2
	b) Explain i) water budget equation ii) Normal ratio method	8	L2	1	1
Unit – II					
4.	a) Discuss the ill effects of over-irrigation on soil and crop productivity.	8	L2	3	1
	b) Determine the weekly evaporation from a reservoir from the following data during the week. Average inflow into the reservoir =32.5 m ³ /s, Average outflow from the reservoir =40.2 m ³ /s, Rainfall during the week =73.6 mm. Average water spread area =15.8 km ² , Seepage =0.25Mm ³ Storage in the beginning of the week = 9180 ham Storage at the end of the week = 8630 ham.	8	L3	4	2

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| a) Determine the dimensions of a canal to carry a discharge of 14 cumec using Kennedy's theory. The side slope of the canal is 0.5:1, Ratio of bottom width to depth is 5.7. | 8 | L3 | 4 | 2 |
| b) Write short notes on:
a) Crop coefficient (K_c)
b) Effective rainfall | 8 | L2 | 3 | 1 |
| a) Explain the design principles of irrigation channels in alluvial soils using Lacey's method. | 8 | L2 | 3 | 2 |
| b) Describe the functions of canal regulators with neat sketches. | 8 | L2 | 4 | 2 |

Unit – III

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|---|---|----|---|---|
| a) Write notes on the importance of site investigation for reservoirs. | 8 | L3 | 5 | 1 |
| b) Explain the steps in stability analysis of a gravity dam by analytical method. | 8 | L2 | 5 | 2 |
| a) Discuss the various modes of failure of gravity dams. | 8 | L2 | 5 | 1 |
| b) Explain the advantages and limitations of earthen dams. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
